

Islamic University of Technology

Department of Software Engineering

Software Project Lab II (SWE 4404)

Project title

AmarShohor

(A Community-Driven Urban Issue Mapping Platform)

Project Proposal Report

Software Project Lab II

Supervisor

Yousuf Ibne kamal Niloy

Group: 02

Team Members

Procheta Silvie (230042114)

Tasnia Farzana (230042124)

Aksan Anan Ria(230042154)

Contents

1 Project Overview **3**

2 Motivation **3**

3 User Requirement Analysis **4**

3.1 Stakeholder Identification 4

3.2 Functional Requirements 5

3.2.1 **Citizen** 5

3.2.2 **Volunteer** 5

3.2.3 **Admin** 5

3.3 Non-Functional Requirements 6

4 System Design **6**

4.1 Issue Lifecycle Flowchart 6

4.2 Class Diagram 8

5 Key Features and Use-Case Descriptions **8**

5.1 Feature 1: Geolocated Issue Reporting 8

5.2 Feature 2: Community Voting and Validation 9

5.3 Feature 3: Live City Map with Heat Zones 9

5.4 Feature 4: Issue Lifecycle Management 9

5.5 Feature 5: Emergency and Crisis Reporting 9

5.6 Feature 6: Role-Based Dashboard System 10

5.7 Feature 7: HelpYourCity – Civic Community Module 10

5.8 Feature 8: Discussion Forums per Issue 10

6 Tools and Technology **11**

7 Proposed Timeline **12**

8 Action Taken Based on Feedback **12**

1. Project Overview

AmarShohor (meaning *"My City"* in Bengali) is a community-driven, web-based urban issue mapping platform designed for Bangladeshi cities. The platform empowers citizens to report, verify, and track urban problems—such as potholes, garbage accumulation, dengue outbreaks, fuel shortages, and load shedding—directly on an interactive live city map. Beyond simple issue reporting, AmarShohor integrates crisis and emergency response features, a community verification system, and a structured volunteer engagement module, creating a unified civic technology ecosystem.

The platform operates through three user roles—**Citizen**, **Volunteer**, and **Admin**—each with clearly defined capabilities and responsibilities. Issues submitted by citizens go through a structured lifecycle: from community voting for validation, through administrative review and approval, to final resolution and closure. This transparent process ensures accountability and builds trust between residents and city authorities.

AmarShohor is built on the **MERN stack** (MongoDB, Express.js, React.js, Node.js), enabling a scalable, real-time web application that can serve urban communities across Bangladesh.

2. Motivation

Urban life in Bangladesh—particularly in Dhaka—presents a daily struggle with infrastructure failures, health crises, and environmental hazards. Despite widespread smartphone access and social media usage, there remains a critical absence of structured, location-aware reporting tools accessible to ordinary citizens.

Several concrete gaps motivate the development of AmarShohor:

- **No centralized reporting platform.** Citizens currently rely on informal social media posts to flag urban problems. These reports get buried under unrelated content and never reach the relevant authorities in a structured format.
- **No resolution tracking.** Once a citizen reports a pothole or a flooded road, there is no mechanism to follow up on whether the problem was addressed. This breeds civic apathy.
- **No structured community participation.** Active citizens who wish to contribute—as volunteers, blood donors, or fundraisers—have no organized channel to do so effectively.
- **Spatial data gaps in crisis situations.** Two recent events starkly illustrate this problem:

- *Measles Outbreak (2026):* The outbreak spread across 58 of 64 districts in Bangladesh, with 166 suspected deaths and vaccine shortages reported. Despite the geographic scale, there was no localized risk-awareness platform to inform residents of high-risk areas near them.
- *Fuel Shortage:* Long queues formed at fuel stations across the country, yet citizens had no real-time, location-based information about which stations had fuel or which areas were critically affected.

AmarShohor directly addresses these gaps by providing a live, community-verified, geolocated platform for urban issue management, emergency reporting, and civic action.

3. User Requirement Analysis

3.1. Stakeholder Identification

The following real-life and representative stakeholders were consulted and analyzed during the requirements phase:

Real-Life Stakeholder: Mr. Rafiqul Islam, Ward Resident – Dhaka North City Corporation

Mr. Rafiqul Islam is a 42-year-old small business owner residing in Ward 18 of Dhaka North City Corporation. He commutes daily through streets that regularly flood during rain and are riddled with unrepaired potholes. He has attempted to report these issues through phone calls to the ward office but received no follow-up. When interviewed, he stated: *"We post on Facebook groups but nothing happens. We need a place where our complaint is registered and we can actually see if it was fixed."* His requirements center on easy mobile reporting, status tracking, and trust in the process.

Additional stakeholders include:

Stakeholder	Type	Primary Need
City residents (general)	Primary	Report and track urban issues
Volunteers / NGO workers	Primary	Browse and respond to issues
City Corporation Admin staff	Primary	Review, approve, escalate issues
Local health authorities	Secondary	Monitor disease outbreak hotspots
Emergency services	Secondary	Receive geo-tagged emergency alerts
Software developers (team)	Internal	Build and maintain the platform

3.2. Functional Requirements

3.2.1. ***Citizen***

- Register and log in with a verified mobile number or email.
- Submit an issue with category selection (infrastructure, health, environment, etc.), GPS pin, photo evidence, and description.
- Track the live status of submitted reports (Pending → Validated → In Progress → Resolved).
- Receive push notifications when a report changes status.
- Submit emergency crisis reports (fire, flood, accident, outbreak).
- Upvote or flag other citizens' reports to assist community validation.
- Participate in issue-specific discussion forums once an issue is validated.
- Access the **HelpYourCity** module to donate blood, food, clothing, or contribute to fundraising campaigns.

3.2.2. ***Volunteer***

- Upgrade citizen account to volunteer status through the platform.
- Browse reported and validated issues within their geographic zone.
- Select and accept tasks aligned with their skills and availability.
- Respond to blood donation match requests.
- Assist in food distribution and emergency crisis response through coordinated tasks.

3.2.3. ***Admin***

- Log in to a role-specific administrative dashboard.
- View issues that are automatically selected through the community voting system.
- Remove false, duplicate, spam, or inappropriate issues when they are reported by users.
- Monitor emergency alerts and high-risk zones using the HeatMap system.
- Re-categorize or re-prioritize validated issues based on urgency and available resources.
- Coordinate with volunteers, NGOs, or city authorities for issue resolution.

- Monitor volunteer activity and identify coverage gaps across city zones.

3.3. Non-Functional Requirements

Category	Requirement
Performance	Pages must load within 2 seconds under normal network conditions.
Scalability	System must support concurrent users from multiple city zones.
Usability	Mobile-first responsive UI accessible to users with minimal digital literacy.
Reliability	Emergency reporting must function with $\geq$99.5% uptime.
Security	JWT-based authentication; role-based access control enforced at API level.
Localization	UI must support both English and Bangla text.

4. System Design

4.1. Issue Lifecycle Flowchart

The flowchart in Figure 1 describes the end-to-end lifecycle of a reported issue within AmarShohor.

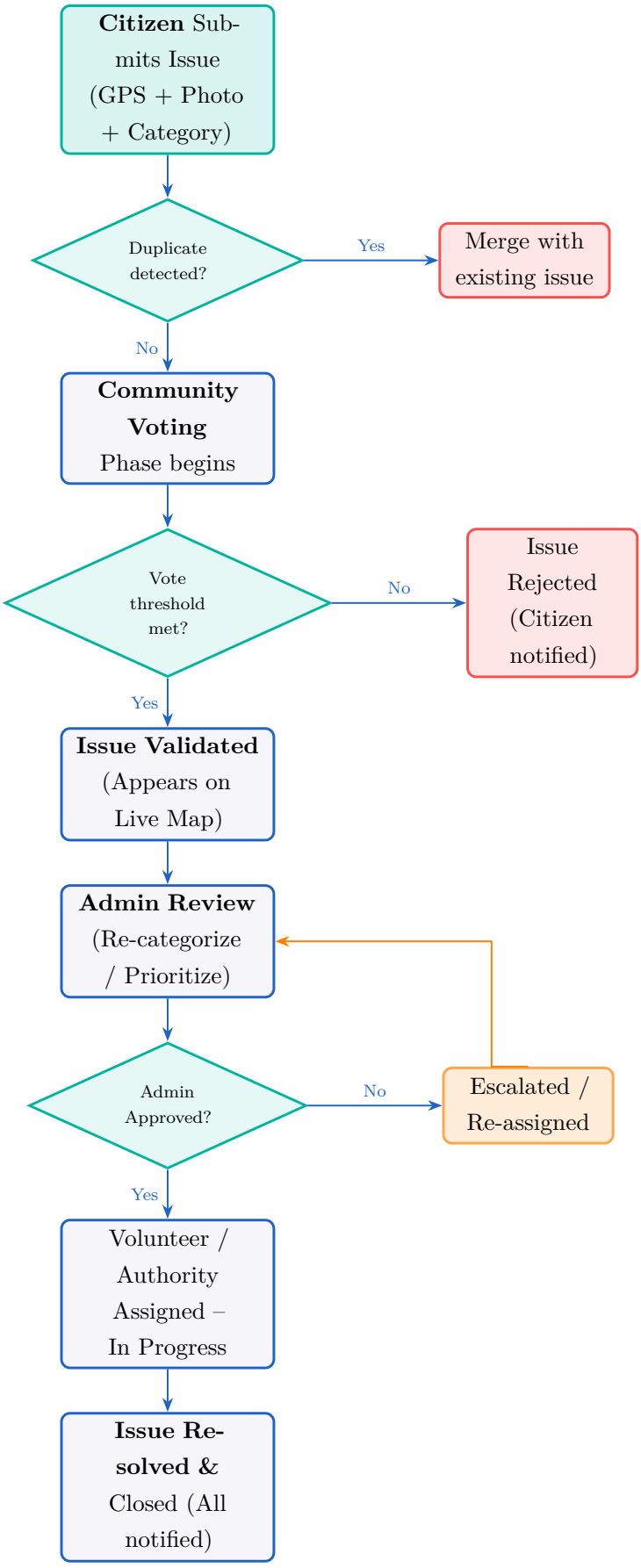


Figure 1: Issue Lifecycle Flowchart in AmarShohor

4.2. Class Diagram

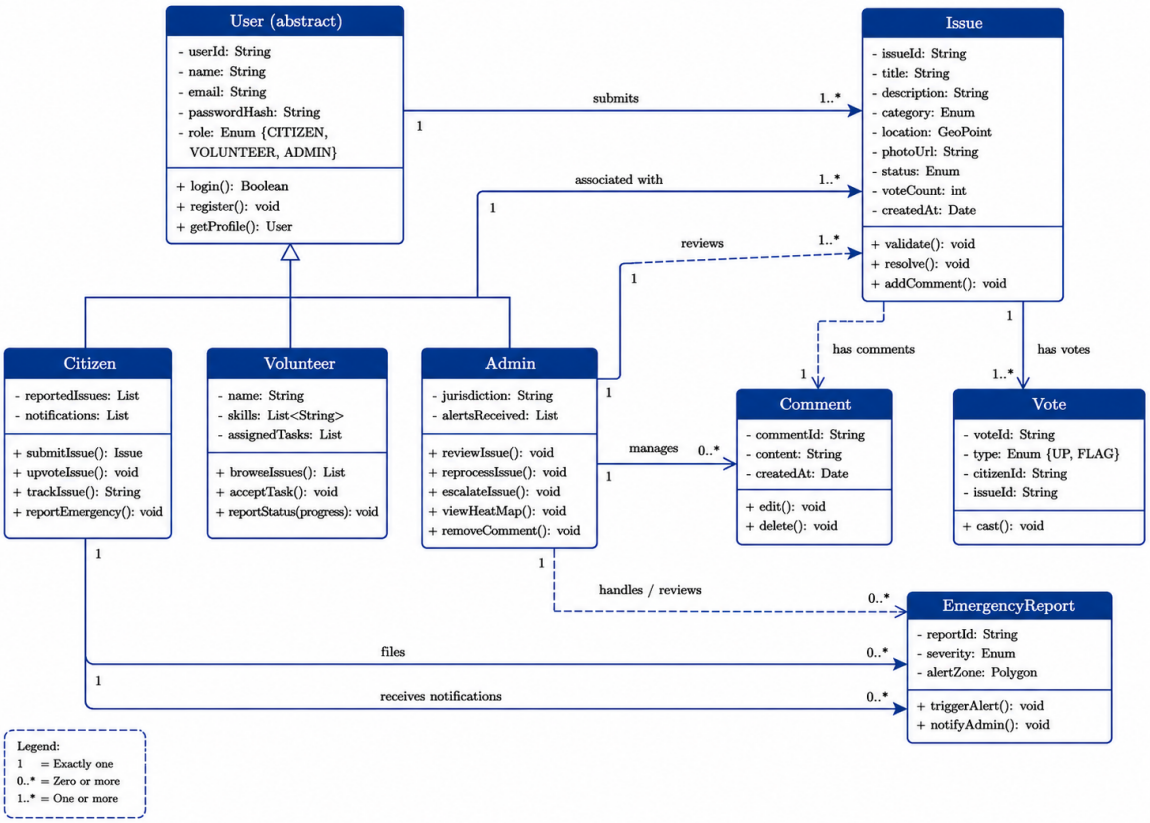


Figure 2: Simplified Class Diagram of AmarShohor

5. Key Features and Use-Case Descriptions

5.1. Feature 1: Geolocated Issue Reporting

Use Case: Submit an Urban Issue

Actor:	Citizen
Precondition:	Citizen is logged in.
Main Flow:	(1) Citizen taps “Report Issue” on the map. (2) System captures GPS coordinates or citizen places a pin manually. (3) Citizen selects issue category (Pothole, Garbage, Health, Fuel, etc.). (4) Citizen uploads a photo and writes a brief description. (5) System checks for duplicate reports within a radius. (6) If no duplicate, issue is created with status *Pending*.
Postcondition:	Issue appears on the community voting queue.

5.2. Feature 2: Community Voting and Validation

Once a citizen submits an issue, other citizens in the area can upvote or flag it. When the upvote count crosses a configurable threshold, the issue is automatically promoted to *Validated* status. This crowd-sourced verification mechanism prevents spam and ensures only genuine issues reach administrative dashboards. Validated issues become visible as pins on the Live City Map.

5.3. Feature 3: Live City Map with Heat Zones

AmarShohor's core interface is an interactive map (rendered via a mapping API) that displays all validated issues as color-coded pins by category. A **HeatMap** layer overlays issue density data, enabling both citizens and administrators to instantly identify high-risk zones. The HeatMap is updated in real time as new issues are validated or resolved.

5.4. Feature 4: Issue Lifecycle Management

Every issue progresses through six defined stages (see Figure 1):

Stage	Status	Description
1	Pending	Submitted, awaiting community votes
2	Voting	Community verification in progress
3	Validated	Vote threshold met; visible on map
4	Under Review	Admin has taken up the issue
5	In Progress	Approved; volunteer/authority assigned
6	Resolved	Work complete; issue closed

Citizens receive notifications at each stage transition. Admins can re-categorize or re-prioritize issues at Stage 4.

5.5. Feature 5: Emergency and Crisis Reporting

Citizens can submit a dedicated **Emergency Report** for time-critical events such as fires, floods, accidents, or disease outbreaks. These bypass the community voting phase and trigger an immediate alert to all city administrators. The system marks the report location with a high-priority pin on the admin dashboard. This feature is designed to complement (not replace) formal emergency services.

5.6. Feature 6: Role-Based Dashboard System

Each user role sees a tailored dashboard:

- **Citizen Dashboard:** My Reports, Area Map, Notifications, HelpYourCity
- **Volunteer Dashboard:** Browse Issues by Zone, My Tasks, Blood Match Requests
- **Admin Dashboard:** Validated Issues Queue, HeatMap Analytics, Emergency Alerts, Volunteer Performance

5.7. Feature 7: HelpYourCity – Civic Community Module

This module transforms AmarShohor from a complaint portal into a proactive civic platform. Citizens can:

- Register as **blood donors** and respond to location-based donation requests.
- Donate **food and clothing** to registered collection drives.
- Launch or contribute to **local fundraising campaigns** for community initiatives.

Volunteers coordinate the on-ground execution of these drives through their dashboard.

5.8. Feature 8: Discussion Forums per Issue

Once an issue reaches *Validated* status, a structured discussion thread is opened for that issue. Citizens can comment, share updates, post photos of progress, and reply to one another. This keeps community awareness alive throughout the resolution process and reduces redundant reports.

6. Tools and Technology

Category	Tool / Technology	Purpose
Frontend	React.js	Component-based, responsive single-page application
	Tailwind CSS	Utility-first styling for mobile-first UI
Backend	Node.js	JavaScript runtime for server-side logic
	Express.js	RESTful API framework; middleware handling
Database	MongoDB	NoSQL document store for flexible issue/user data
Mapping	Leaflet.js	Open-source interactive map rendering
	OpenStreetMap	Base tile provider for city maps
Auth & Security	JSON Web Tokens	Stateless authentication
	bcrypt	Password hashing
Realtime	Socket.IO	Live notifications and map pin updates
	REST API	Standard CRUD operations
DevOps	GitHub	Version control, collaborative development

The MERN stack was chosen for its JavaScript uniformity across the entire application, strong community support, and suitability for real-time, data-intensive web applications. MongoDB's flexible schema is particularly well-suited to the varied structure of issue reports across different categories.

7. Proposed Timeline

The project is planned over **14 weeks**, organized into overlapping development sprints as described in Figure 3 and visualised in the Gantt chart below.

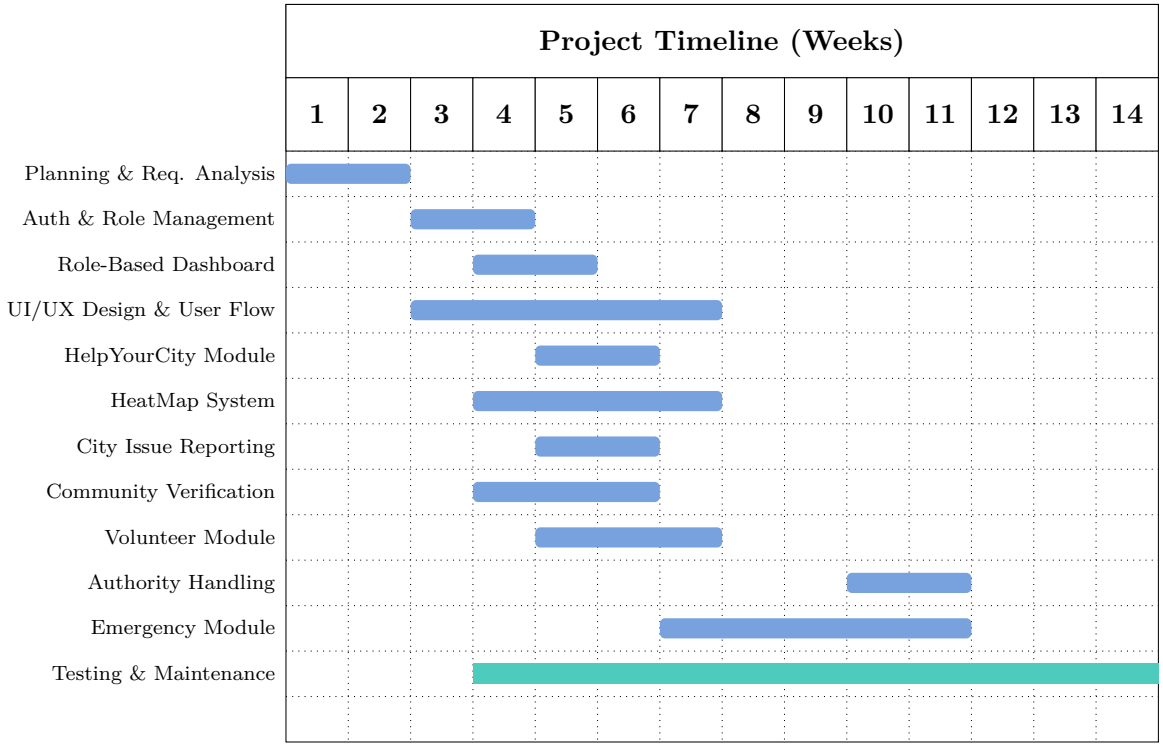


Figure 3: Gantt Chart – AmarShohor Development Schedule

8. Action Taken Based on Feedback

Following the initial proposal presentation, the team received several pieces of feedback from the supervising faculty. The following actions were taken in response:

1. **Clarification of the Issue Lifecycle.**
   The original proposal described the issue flow informally. Following feedback on the need for a structured process, the team formalized a six-stage Issue Lifecycle (Citizen Reports → Community Voting → Issue Validated → Admin Review → Admin Approval → Issue Resolved & Closed) and incorporated it as a core architectural element.

2. **Scope Refinement – Admin Module.**
   An early version of the proposal needed the Admin to accept and validate the issues after being verified with public votes. Based on feedback, now Admin does not need to accept the issues manually. They are chosen automatically by users vote. However if users report one issue they can remove it.

Conclusion

AmarShohor represents a meaningful step towards digitizing civic participation in Bangladesh. By combining geolocated issue reporting, community verification, live mapping, emergency alerting, and volunteer coordination into a single, accessible platform, it addresses a persistent gap between citizens and urban authorities. The MERN-based architecture ensures the platform is scalable and maintainable, while the carefully designed role system keeps interactions structured and accountable. The team is committed to delivering a fully functional prototype within the proposed 14-week timeline.